TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 13

a. 2 C=O bonds and 4 C-H bonds on each side of the equation

c. $\Delta H^{0}_{\text{isodesm}}$ = 40.143599 + 187.617929 - 2 · 113.833332 = 0.094864 hartree = 249 kJ/mol

d. -109.6 kJ/mol

The value is in good agreement with the experimental value of -115.9 kJ/mol.

Supplementary Exercise

The enthalpies of formation using the results of Exercise 12 are:

Enthalpy of formation = 1403.4 kJ/mol - Enthalpy of atomization

Method	Enthalpy of atomization [kJ/mol]	Enthalpy of formation [kJ/mol]
HF/6-31G(d)-isodesmic	---	-110
HF/6-31G(d)-atomization	1002	402
MP2/6-31G(d)	1409	-6

SVWN5/6-31G(d)	1763	-360
Becke3LYP/6-31G(d)	1501	-98
G1	1513	-110
experimental	1519	-116

[General help] [To overview] [Compute] [Contact the course supervisor]

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